

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

**Hybrid Ensemble Machine Learning with Explainable AI
for Early Prediction of Parkinson's Disease Severity**

submitted in partial fulfilment of the requirements for the award of degree of

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In

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled "Hybrid Ensemble Machine Learning with Explainable AI for Early Prediction of Parkinson's Disease Severity" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Ajay Katiyar. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

Parkinson's disease is a progressive brain disorder that affects the movement and speech of patients, leading to difficulties in daily activities and communication. It has affected more than 10 million patients worldwide. Doctors use a score called UPDRS (Unified Parkinson's Disease Rating Scale) to measure how severe this disease is, and going to the hospital every time makes it difficult and expensive for the patient, making voice a practical alternative for remote monitoring.

Most existing machine learning studies on this problem suffer a critical flaw known as data leakage, where voice recordings from the same patient appear in both training and testing sets. This does not reflect real-world performance on new patients. This study proposes a subject-independent hybrid stacking ensemble that combines four regression models: Random Forest, XGBoost, Support Vector Regression, and Multilayer Perceptron. Subject-wise GroupKFold cross-validation generates out-of-fold predictions at the stacking level, avoiding patient-level data leakage.

Experiments are conducted on the Oxford Parkinson's Disease Telemonitoring dataset comprising 5,875 voice recordings from 42 subjects. Results demonstrate that XGBoost achieves $R^2 = 0.8767$ and $MAE = 2.34$ UPDRS points on a hold-out cohort of nine completely unseen patients. The stacking ensemble records $R^2 = 0.7346$. Wilcoxon signed-rank testing confirms all performance differences are statistically significant ($p < 0.05$). SHAP analysis identifies that Parkinson's severity is mostly predicted by the motor subscore, patient age, Detrended Fluctuation Analysis (DFA), and Pitch Period Entropy (PPE).

Keywords: Parkinson's disease, XGBoost, stacking ensemble, UPDRS prediction, SHAP explainability, machine learning, GroupKFold, data leakage.

1. Introduction

Parkinson's disease (PD) is the second most widespread neurological disorder, impacting over ten million people globally, exerting a major load on patients, families, and the healthcare system. It is characterized by progressive motor impairment — including tremor at rest, muscle rigidity, bradykinesia, and postural instability — in addition to non-motor symptoms such as speech impairment, autonomic dysfunction, and cognitive change. Clinical severity is formally measured using the Unified Parkinson's Disease Rating Scale (UPDRS), which is periodically administered by trained neurologists and assigns number scores across motor and non-motor domains.

One of the main practical issues with UPDRS-based monitoring is the need for in-person clinical appointments. Visiting a speciality facility is expensive and time-consuming for patients with severe mobility impairment, and because these trips are rare, daily variations in symptom severity are never recorded. Vocal biomarkers derived from sustained phonation recordings offer an appealing substitute: the laryngeal structure and speech motor control are gradually weakened by Parkinson-related neurodegeneration, resulting in quantifiable alterations in nonlinear voice dynamics, amplitude stability, and pitch regularity that may be recorded remotely using standard microphones.

However, the work already done on this dataset contains a recurring methodological flaw. The great majority of published research separated the 5,875 recordings into training and test sets at the record level rather than at the patient level. Because each of the 42 patients provided approximately 140 recordings, random splitting ensures that recordings from the same individual appear in both divisions. This results in a form of data leakage that inflates performance metrics significantly. A second problem is the lack of model transparency — for any diagnostic algorithm to be used in clinical practice, clinicians need to understand the reasoning behind each prediction.

1.1 Objectives

- Implement a patient-level, subject-independent 80/20 train-test split ensuring completely unseen test patients
- Develop a two-level hybrid stacking ensemble that avoids patient-level leakage at both base-learner and meta-learning stages

- Apply SHAP-based explainability to the trained XGBoost model for clinical interpretability
- Verify all reported model performance differences using Wilcoxon signed-rank statistical tests

1.2 Problem Statement

Most existing machine learning studies for UPDRS prediction from voice biomarkers suffer from data leakage due to record-level rather than patient-level train-test splitting. This inflates performance metrics and does not reflect real-world generalization to new patients. Additionally, black-box model predictions are unsuitable for clinical adoption without interpretability tools. This project addresses both these issues by proposing a rigorous subject-independent evaluation framework paired with SHAP explainability.

1.3 Scope of the Project

This project focuses on remote monitoring of Parkinson's disease severity using vocal biomarkers from the publicly available Oxford Parkinson's Disease Telemonitoring dataset. The scope includes dataset preprocessing, subject-independent train-test split design, training and evaluation of four machine learning regressors, development of a hybrid stacking ensemble, SHAP-based feature attribution analysis, and statistical significance testing.

2. Methodology

2.1 Dataset Description

The Oxford Parkinson’s Disease Telemonitoring dataset, publicly accessible via the UCI Machine Learning Repository, is used in this study. The Intel Corporation telemonitoring platform collected data from 42 patients diagnosed with Parkinson’s disease in its early stages. Over a six-month monitoring period, each patient repeatedly recorded a vowel phonation using a home microphone. The full dataset contains 5,875 recordings with no missing values or duplicate entries, averaging approximately 140 observations per subject.

Among the 22 raw properties, each record includes a patient subject identity, age, sex, and a test_time field indicating days since enrolment. Two clinical outcome scores are provided: total_UPDRS (the regression target) and motor_UPDRS (the motor sub-component). Acoustic features comprise the remaining 16 attributes: two noise ratio measures (NHR and HNR), three nonlinear complexity measures (RPDE, DFA and PPE), five Jitter variations quantifying pitch disruption, and six Shimmer variants quantifying amplitude perturbation.

Table I presents a comparison of prior work on UPDRS prediction from voice biomarkers, highlighting the methodological gaps — particularly regarding data leakage — that motivated the subject-independent framework adopted in this study.

Study	Method	Target	Best R ²	Split Strategy	Leakage Risk
Tsanas et al. (2010) [3]	CART / LASSO	Total UPDRS	0.8320	Record-level random	High
Grover et al. (2018) [4]	Deep Learning	Total UPDRS	0.9100	Record-level random	High
Ali et al. (2019) [5]	LDA + Neural Net	Total UPDRS	0.8850	Record-level random	High
Pahuja & Bhatt (2021) [9]	SVM / RF / KNN	Total UPDRS	0.8720	Record-level random	High
Abdellatif et al. (2023) [10]	Stacking Ensemble	Total UPDRS	0.8940	Record-level random	High
This Study (XGBoost)	XGBoost + SHAP	Total UPDRS	0.8767	Subject-level (GroupKFold)	None

Table I. Comparison of Prior Work on UPDRS Prediction from Voice Biomarkers

2.2 Preprocessing

The `test_time` feature was eliminated because it encodes longitudinal position within the follow-up sequence, information unavailable for a new patient at prediction time. The `motor_UPDRS` attribute was retained as a predictor since the same clinician records both motor and total UPDRS scores simultaneously. Twenty predictor features — age, sex, `motor_UPDRS`, and the 16 acoustic biomarkers — are used to describe each observation following preprocessing.

2.3 Subject-Independent Data Partitioning

All 1,235 recordings from 9 patients make up the hold-out test group, whereas the remaining 4,640 recordings from 33 patients form the training set, according to a subject-level 80/20 division with random seed set at 42. No recording from any test patient was exposed to the model during training. This approach directly prevents the leakage reported in most previous studies, where record-level random splits allow the model to observe the same patient’s speech in both training and evaluation phases.

2.4 Base Learners

Four regression models were trained and assessed as base learners in the stacking ensemble as well as independent predictors:

- Random Forest: Builds 200 decision trees using bootstrap samples and random feature subsets. Trained on unscaled inputs, invariant to monotonic feature scaling.
- XGBoost: Constructs sequential gradient-boosted trees with `n_estimators=300`, `learning_rate=0.05`, and `max_depth=5`. Integrated L1/L2 regularization controls inter-correlation between Jitter and Shimmer feature families.
- Support Vector Regression (SVR): Uses radial basis function kernel with `C=1.0`, integrated into a scikit-learn pipeline applying `StandardScaler` fitted only on training data.
- Multilayer Perceptron (MLP): Hidden layers of size (128, 64), ReLU activations, and Adam optimizer for up to 1,000 iterations, with analogous scaling pipeline.

2.5 Subject-Wise GroupKFold Cross-Validation

Five-fold cross-validation was performed throughout using scikit-learn’s `GroupKFold` with subject identifiers as group labels, so that every validation fold consists exclusively of patients

absent from that fold's training partition. This arrangement prevents the same patient's vocal characteristics from appearing in both the training and validation halves of any fold. Standard KFold, which ignores patient identity when constructing folds, was explicitly avoided.

2.6 Hybrid Stacking Ensemble

The stacking ensemble operates at two levels. Using the subject-wise GroupKFold process, each of the four base learners at the first level produces out-of-fold (OOF) predictions for the 4,640 training recordings. Folds are patient-disjoint, so predictions for any training recording are always made by a model that has never seen another recording from the same patient. A $4,640 \times 4$ meta-feature matrix is created by assembling the four OOF prediction vectors.

At the second level, an XGBoost meta-learner (`n_estimators=200`, `learning_rate=0.05`, `max_depth=3`) is trained on this meta-feature matrix. All base learners are retrained on the whole training set for test-set inference; their predictions on the 1,235 test recordings create a $1,235 \times 4$ meta-test matrix from which the meta-learner generates final predictions.

2.7 SHAP Explainability Analysis

The trained XGBoost model was subjected to post-hoc feature attribution using TreeExplainer from the SHAP library. SHAP calculates an exact attribution value for each of the 1,235 test observations, indicating each feature's marginal contribution to the prediction's departure from the model's overall mean output. Positive SHAP values push the prediction above the average; negative values push it below. Mean absolute SHAP values determine feature importance rankings.

2.8 Evaluation Metrics and Statistical Testing

Three regression metrics are reported: R^2 (coefficient of determination), MAE (Mean Absolute Error), and RMSE (Root Mean Squared Error). The Wilcoxon signed-rank test — a non-parametric paired technique — was employed to determine if the absolute per-sample prediction errors of two models on the same test set differ significantly. At $\alpha = 0.05$, the null hypothesis of similar error distributions was rejected.

3. Tools and Technologies

3.1 Programming Language

- Python 3.10: Primary programming language used for data processing, model development, and analysis.

3.2 Machine Learning Libraries

- scikit-learn: Used for SVR, Random Forest, MLP, GroupKFold cross-validation, StandardScaler, and performance metric computation.
- XGBoost: Gradient-boosted tree implementation used for both base learner and meta-learner in the stacking ensemble.
- SHAP (SHapley Additive exPlanations): Post-hoc model explainability library used with TreeExplainer for feature attribution on the XGBoost model.

3.3 Data Handling

- NumPy: Numerical computation and array operations.
- Pandas: Dataset loading, preprocessing, and feature engineering.
- UCI Machine Learning Repository: Source of the Oxford Parkinson's Disease Telemonitoring dataset (5,875 recordings from 42 subjects).

3.4 Statistical Testing

- SciPy: Used for Wilcoxon signed-rank statistical significance testing between model pairs.

3.5 Visualization

- Matplotlib: Used for plotting learning curves, SHAP beeswarm plots, and waterfall decompositions.
- SHAP visualization utilities: Beeswarm plot, waterfall decomposition, and dependence plots for top features.

3.6 Development Environment

- Jupyter Notebook: Interactive development environment for iterative model development and visualization.

- Google Colab: Cloud-based GPU-enabled environment for model training.
- Git/GitHub: Version control and collaboration.

4. Implementation

4.1 Data Loading and Preprocessing

The Oxford Parkinson’s Disease Telemonitoring dataset was loaded from the UCI Machine Learning Repository in CSV format. Initial exploratory analysis confirmed no missing values or duplicate entries across the 5,875 observations. The `test_time` column was dropped to eliminate temporal leakage. The `subject` column was retained solely for stratified splitting purposes. The remaining 20 features (age, sex, motor_UPDRS, and 16 acoustic biomarkers) were retained as predictors, with `total_UPDRS` as the regression target.

4.2 Subject-Independent Data Splitting

Unique subject identifiers were extracted and shuffled with random seed 42. The first 33 subjects were assigned to training (4,640 recordings) and the remaining 9 subjects to testing (1,235 recordings). This patient-level split ensured no recordings from test subjects were seen during model training, directly preventing the data leakage that affects most prior work on this dataset.

4.3 Base Learner Training

Each of the four base learners was instantiated with fixed hyperparameters and trained on the full training set. For SVR and MLP, scikit-learn Pipelines were used with `StandardScaler` fitted exclusively on training data to prevent test-set statistics from influencing feature normalization. Random Forest and XGBoost were trained on unscaled features. Five-fold subject-wise `GroupKFold` cross-validation with subject identifiers as group labels was applied during cross-validation scoring to prevent fold-level leakage.

4.4 Stacking Ensemble Construction

For each of the five `GroupKFold` folds, each base learner was trained on four folds and used to generate out-of-fold predictions on the held-out fold. These OOF predictions were assembled into a $4,640 \times 4$ meta-feature matrix. An XGBoost meta-learner with `n_estimators=200`, `learning_rate=0.05`, and `max_depth=3` was then trained on this matrix. For test inference, all four base learners were retrained on the complete training set, their predictions on the 1,235 test recordings assembled into a $1,235 \times 4$ meta-test matrix, and the meta-learner applied to generate final ensemble predictions.

4.5 SHAP Analysis

The SHAP TreeExplainer was instantiated with the trained XGBoost model and applied to the 1,235 test observations. Exact Shapley values were computed for each observation and each feature. Feature importance was ranked by mean absolute SHAP value across all test observations. Visualizations generated include: a beeswarm summary plot showing the distribution of SHAP values per feature, dependence plots for the top three features (motor_UPDRS, age, DFA), and a single-patient waterfall decomposition for a representative test case with total_UPDRS = 42.1.

4.6 Statistical Significance Testing

For each pair of models of interest, the absolute per-sample prediction errors on the 1,235 test recordings were extracted. The Wilcoxon signed-rank test from SciPy was applied to each pair, testing the null hypothesis that the error distributions are identical. All p-values were compared against $\alpha = 0.05$ to determine statistical significance.

5. Major Findings / Outcomes / Results

5.1 Individual Model Performance

The results of the subject-independent test cohort for each of the four base learners are shown in Table II. With the highest test R^2 of 0.8767, XGBoost explains almost 88% of the variance in total UPDRS scores for completely unseen patients. The equivalent MAE of 2.34 UPDRS points indicates an average prediction error of about two points on a clinical scale ranging from 7 to 55, which is considered acceptable for ongoing remote monitoring. SVR and Random Forest obtain test R^2 values of 0.7850 and 0.7798 respectively, while MLP performs poorly at $R^2 = 0.6024$.

Model	CV R^2 (Mean $\pm$ Std)	Test R^2	Test MAE	Test RMSE
XGBoost	0.7387 ± 0.1270	0.8767	2.3400	3.1962
SVR	0.7310 ± 0.0765	0.7850	3.2319	4.2210
Random Forest	0.7275 ± 0.1354	0.7798	3.1303	4.2718
MLP	0.7133 ± 0.0762	0.6024	4.4185	5.7401

Table II. Individual Model Performance on Subject-Independent Test Set

5.2 Cross-Validation Fold Analysis

Per-fold validation R^2 results (Table III) reveal significant variation across folds. XGBoost varies by over 0.28 R^2 units, from 0.5967 in Fold 1 to 0.8743 in Fold 2, reflecting genuine patient heterogeneity rather than model instability. Folds 1 and 4 are consistently the hardest across all four models simultaneously, indicating that patients in those folds have voice severity profiles that none of the four algorithms can reliably deduce from the remaining training cases.

Fold	RF R^2	XGB R^2	SVR R^2	MLP R^2
1	0.5661	0.5967	0.8176	0.6454
2	0.8880	0.8743	0.8056	0.8617
3	0.7532	0.7660	0.6283	0.6757
4	0.5755	0.5850	0.6573	0.6869
5	0.8546	0.8717	0.7463	0.6969

Table III. Per-Fold Validation R^2 Across Subject-Wise GroupKFold (5 Folds)

5.3 Stacking Ensemble Results

With $R^2 = 0.7346$, $MAE = 3.4401$ and $RMSE = 4.6894$, the stacking ensemble performs worse than XGBoost, SVR, and Random Forest (Table IV). This finding challenges the widely held belief that ensemble complexity consistently outperforms the best individual model. Three factors explain this result: (1) only 33 subjects' OOF predictions are used to train the meta-learner, limiting signal; (2) all four base learners fail simultaneously on the same patient groups (Folds 1 and 4), preventing the ensemble from correcting collective errors; and (3) XGBoost's high individual accuracy means combining it with weaker models adds noise rather than correcting residual mistakes.

Model	Type	R^2	MAE	RMSE
XGBoost	Individual	0.8767	2.3400	3.1962
SVR	Individual	0.7850	3.2319	4.2210
Random Forest	Individual	0.7798	3.1303	4.2718
Stacking Ensemble	Ensemble	0.7346	3.4401	4.6894
MLP	Individual	0.6024	4.4185	5.7401

Table IV. Stacking Ensemble vs. Individual Models — Subject-Independent Test Set

5.4 Statistical Significance

Every pairwise comparison results in a p-value less than 0.05 (Table V), confirming that random variation cannot account for any of the reported performance differences. XGBoost's dominance over all other models is very pronounced ($p < 0.0001$ in each instance), providing compelling statistical evidence that its success is systematic rather than accidental.

Comparison	p-value	Significant?
XGBoost vs. Random Forest	<0.0001	Yes
XGBoost vs. SVR	<0.0001	Yes
XGBoost vs. MLP	<0.0001	Yes
XGBoost vs. Stacking Ensemble	<0.0001	Yes
Stacking vs. Random Forest	0.0295	Yes
Stacking vs. SVR	0.0224	Yes
Stacking vs. MLP	<0.0001	Yes

Table V. Wilcoxon Signed-Rank Test Results (Significance Level $\alpha = 0.05$)

5.5 SHAP Feature Importance Analysis

The 10 most significant features from the 1,235 test observations are listed in Table VI according to mean absolute SHAP value. motor_UPDRS leads with a mean absolute SHAP score of 8.291 as a clinically co-measured sub-component of the target variable. Among voice-only characteristics, the three nonlinear complexity features — DFA, PPE, and RPDE — consistently outperform classical perturbation features. This aligns with known Parkinson-related dysphonia pathophysiology, where neurodegenerative processes impair the larynx’s fine motor control, resulting in increased irregularity better captured by nonlinear measures.

Rank	Feature	Mean SHAP	Clinical Interpretation
1	motor_UPDRS	8.2911	Motor sub-score — concurrent clinical measurement
2	age	1.1798	Older patients show progressively higher UPDRS severity
3	sex	0.4545	Sex-linked variation in Parkinson’s disease progression
4	DFA	0.3801	Detrended Fluctuation Analysis — vocal fractal irregularity
5	Jitter (Abs)	0.2293	Absolute pitch perturbation — vocal tremor indicator
6	Shimmer: APQ11	0.0775	11-cycle amplitude perturbation — amplitude instability
7	PPE	0.0739	Pitch Period Entropy — voice complexity measure
8	Jitter: RAP	0.0618	Relative Average Perturbation in fundamental frequency
9	HNR	0.0615	Harmonics-to-Noise Ratio — overall voice quality
10	RPDE	0.0515	Recurrence Period Density Entropy — nonlinear vocal dynamics

Table VI. Top Ten Features by Mean Absolute SHAP Value (XGBoost, Test Set)

6. Conclusion and Future Scope

6.1 Conclusion

This project created a machine learning system that forecasts the severity of Parkinson's disease by analyzing a patient's voice recording. The primary objective was to create an honest system — one that is tested on patients it has never encountered before during training.

Four machine learning models were trained and compared: XGBoost, Random Forest, Support Vector Regression, and Multilayer Perceptron. With an accuracy score of $R^2 = 0.8767$ and an average prediction error of only 2.34 UPDRS points, XGBoost outperformed all other models. This indicates that the model can accurately estimate Parkinson's disease severity from voice recordings alone, without requiring a clinical visit. Statistical tests (Wilcoxon signed-rank) confirmed XGBoost's superiority at $p < 0.0001$.

A hybrid stacking ensemble was constructed by combining four models. Despite its complexity, the stacking ensemble did not outperform XGBoost alone ($R^2 = 0.7346$ vs. 0.8767), demonstrating that when patient cohort size is constrained and base models fail synchronously on the same patient groups, ensemble complexity does not provide additional benefit.

SHAP analysis was applied to make the XGBoost model clinically interpretable. The most significant predictors of Parkinson's disease severity are the motor sub-score, patient age, and two voice characteristics — DFA and PPE — enabling clinicians to understand precisely why the model predicted a certain outcome for any given patient.

6.2 Future Scope

- External validation: Test the model on a second dataset from a different country or demographic group to assess generalizability across all patient populations.
- Leave-One-Subject-Out (LOSO) validation: Apply this more rigorous subject-independent testing strategy to further improve the dependability and reproducibility of results.
- Bayesian hyperparameter optimization: Automatically find better XGBoost settings using Bayesian optimization to potentially increase accuracy beyond the current $R^2 = 0.8767$.

- Longitudinal deep learning: Investigate deep learning models such as LSTM networks that can analyze multiple voice recordings of the same patient over time, leveraging the longitudinal structure of the dataset.
- Mobile application integration: Develop a mobile application allowing patients to record their voice at home and receive an instant severity assessment without visiting the hospital, enabling continuous remote monitoring.

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